



Update Meetings

3D Spatial Scene Description for Blind People

Lukas Ederer

Tobias Lewald

Jan Duchscherer

Fakultät für Informatik und Mathematik
Hochschule München

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Update Meetings — Overview

#	Date
1	27.03.2025
2	01.04.2025
3	...

Update Meeting 1 — Tasks Worked On

- Play with Gemini Spatial understanding and get initial idea of its capabilities and limitations.
- Get general understanding of the project goals.
- Implement basic voice to text via OpenAI Whisper (needed for some other task).
- Reflect upon questions for clarification.
- Reflect general ideas for the holistic project.
 - Dynamic Semantic Scene Graph, SLAM, 3D native object detection, etc.

Update Meeting 1 — Problems Encountered

- How does the Data look like? What features are available?
- What models are currently being used?
- Wie sieht die Entwicklungsumgebung aus (Geräte, iPhone notwendig?)
- Entwicklung auf nicht Apple-Geräten.
- What parts of the application should finally run on the iPhone?
- API Costs: Who will pay? → *SmartAls*
- Initial ambiguities w.r.t. project goals.
- Gemini 2.0 flashes 3D OOBs do not seem reliable.

Update Meeting 1 — Plans for Next Week

- Research on one-shot end to end multimodal detection models.
- Await Data from Yamen and do initial exploration.
- Explore data structures (StrayScanner, Record3D)
- Get Gemini 2.0 flash via API running.

Optional:

- Combination of depth data w/ detections.
- Grobe Überlegungen zur Modellarchitektur.
- Explore potential network architectures.

Update Meeting 2

[Date]

Update Meeting 2 — Tasks Worked On

- **SpatialBot**: VLM fine-tuned on **RGBD** data for better 3D understanding [?].
- **SpatialBench** for evaluating VLMs spatial understanding.

Update Meeting 2 — Problems Encountered

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Update Meeting 2 — Plans for Next Week

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